

MEHMET SEMIH BABACAN

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Software Engineer, and Product Manager with 3+ years of experience building and scaling **customer-driven, B2C, AI-powered products**. With dual degrees in **Computer Science and Industrial Engineering**, I bridge **technical execution with product and business strategy**—owning work from **problem discovery and system design to delivery and iteration**. I lead **AI product strategy and execution**, including **LLM- and agent-based architectures, roadmap definition, and cross-functional alignment** across engineering, design, and leadership. Experienced in **full-stack development, systems design, and data-informed decision-making**, I thrive in fast-paced environments where **turning complex problems into scalable, real-world impact** is the goal.

EDUCATION & RESEARCH

B.Sc. Industrial Engineering, Yildiz Technical University — GPA: 3.32 2018–2024

B.Sc. Computer Science (Double Major), Yildiz Technical University — GPA: 3.05 2020–2024

Publication: [Beyond Synthetic Benchmarks: Assessing Recent LLMs for Code Generation](#)

AI Product Manager

Upily

Aug. 2025 – Present

Remote / Global

- Own the AI product strategy for multiple B2C applications, translating company goals into executable AI roadmaps and measurable outcomes. Integrating supportive AI based and agentic features for Wonjo Kids and other apps.
- Define and lead the end-to-end AI architecture, including LLM selection, agent design, orchestration patterns, and production deployment standards.
- Deliver AI products from concept to scale: research and prototyping, model and agent integration, and production rollout. Drive alignment across engineering, design, and leadership to ensure scalable, user-centric AI systems.

Head of Product / Co-Founder

Solace Technology Incorporated

Nov. 2023 – Aug. 2025

Istanbul, Turkey

- Launched two AI-native products—**Ensi** (smart home) and **EMA** (B2B assistant)—driving full-stack development, AI integration, product strategy, and user-centric iteration.
- Built scalable backend and agent systems (Python/FastAPI, WebSockets, LangChain/LangGraph, RAG, STT/TTS) and internal Next.js tools; managed Firebase, PostgreSQL, SQLite, and Supabase infrastructure.
- Implemented automated CI/CD pipelines (GitHub Actions, Docker, Nginx) enabling rapid releases and stable agentic workflows across mobile, web, and backend platforms.
- Raised **\$250K**, secured TÜBİTAK 1507 grant, joined **Bilişim Vadisi**, and operated with a lean **\$200K** budget while pivoting from hardware to SaaS through user discovery and rapid prototyping. Grew team from **4 to 8**, led engineering and AI execution, and handled GTM, brand strategy, launch campaigns, demos, operations, legal, and investor relations.

Part-Time Researcher

Scientific and Technological Research Council of Turkey (TÜBİTAK)

Aug. 2023 – Dec. 2023

Kocaeli, Turkey

- Built a secure cross-platform MPC Wallet in Flutter+Go to eliminate single-key custody risk.
- Contributed to a Go crypto library and simulated adversarial attacks to test protocol resilience.

STAR Presidential Research Intern

Scientific and Technological Research Council of Turkey (TÜBİTAK BILGEM)

Jan. 2022 – Jul. 2022

Gebze, Turkey

- Built CBDC smart contracts with Hyperledger Fabric, Solidity, and TypeScript for the Central Bank.
- Developed and deployed a regulated NFT prototype to test digital asset flows internally.

Industrial Engineering Internships (3)

2020 – 2023

Istanbul & Kocaeli, Turkey

- **Torun Metal (ERP Intern)** — Supported **SAP MM/PP** rollout by gathering shop-floor data, generating live **KPI dashboards** (throughput, availability), and assisting inventory analysis using SQL/Excel.
- **Baykar Defense / Cezeri AI (Project Management Intern)** — Benchmarked CI/CD tools, built containerized pipelines, and proposed a **DevOps roadmap** including early-stage static analysis and deployment KPIs.
- **KGM Heavy Machinery (Production Intern)** — Conducted time studies, lean layout improvements, and **REBA/RULA ergonomic assessments**, delivering a factory report with **OEE** and resource planning insights.

UNIVERSITY CLUB & VOLUNTARY WORK

Deep Learning Engineer, Autonomous Systems Club

Nov. 2020 – Aug. 2021

Yildiz Technical University

Istanbul, Turkey

- Boosted detection pipeline throughput (YOLOv4 → TensorRT/TKDNN) from 36 to 140 FPS. Led dataset ops and launched Tableau reviews to ensure labeling and evaluation quality.

Rising Star Scholar, Turkish Technology Team Foundation

Feb. 2021 – Nov. 2022

Istanbul, Turkey

- Built a desktop evaluation system and visualized 600K+ records to aid the human resources operations. Mentored across AI, software, and entrepreneurship; supported TEKNOFEST operations.

TECHNICAL SKILLS & COMPETENCIES

GenAI / Agents	Cursor, Windsurf, LangChain, n8n, OpenAI, Claude
Backend	Python, FastAPI, Node.js, REST, Firebase, Supabase
DevOps & Infrastructure	Docker, GitHub Actions, CI/CD, GCP, Nginx
Frontend	React, Next.js, Flutter, Sanity CMS, HTML/CSS
Mobile / Desktop Apps	Flutter, React Native, PyQt, C++ Qt
Programming Languages	Python, TypeScript, JavaScript, Go, SQL, Bash, Dart, C/C++
Machine Learning	PyTorch, TensorFlow, Scikit-learn, XGBoost
Data & Visualization	Pandas, Polars, NumPy, Matplotlib, Tableau
Blockchain	Solidity, Circom, Remix IDE, Go
Dev Tools & Databases	Vim, VS Code, Copilot, PostgreSQL, MongoDB

Soft Skills & Competencies: Analysis, Agility, Problem Solving, Leadership, Systems Thinking, Tech Trends, Code Quality, UX Empathy, Drive, Optimization, Mentoring, Languages.

CERTIFICATES

Google	Project Management Specialization; Game & App Dev Academy
DeepLearning.AI	Deep Learning Specialization
Codecademy	R, Python, Data Visualization, Data Analysis
Mennan Şahin (TRT)	Diction & Effective Speaking in Turkish

LANGUAGES

English (Full Professional)

Turkish (Native)

Spanish (Intermediate)